

# Intro to Counting

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# Overview

## Counting Sequences (15.2)

- The Product Rule (15.2.1)

- Subsets of an  $n$ -element Set (15.2.2)

- The Sum Rule (15.2.3)

## The Generalized Product Rule (15.3)

- Defective Dollar Bills (15.3.1)

- A Chess Problem (15.3.2)

- Permutations (15.3.3)

# Continuous vs. Discrete

| continuous           | discrete         |
|----------------------|------------------|
| sounds               | words            |
| clay                 | legos            |
| photographs          | diagrams         |
| mouse                | keyboard         |
| sitting on the floor | sitting in seats |
| $\mathbb{R}$         | $\mathbb{Z}$     |
| quantities           | numbers          |
| measuring            | <b>counting</b>  |

## General strategy for counting

- ▶ Get good at counting some categories of things.
- ▶ Use bijections to relate one of those to the problem at hand.
- ▶ The main category we will study is *counting sequences*.

# Product rule

**Rule:** If  $P_1, P_2, \dots, P_n$  are finite sets, then:

$$|P_1 \times P_2 \times \cdots \times P_n| = |P_1| \cdot |P_2| \cdot \cdots \cdot |P_n|.$$

Example:

- ▶ destination = { Mars, Moon, Venus }
- ▶ nationality = { US, Russia, China, India }
- ▶ robotic = { yes, no }

If a space trip consists of a choice of destination (one of 3), nationality (one of 4), and whether it is robotic (one of 2), how many different space trips are there?  $3 \times 4 \times 2 = 24$ .

# Freestyle



## Number of subsets rule

We showed that the number of subsets of a set of size  $n$  has a bijection with the set of binary strings of length  $n$ .

Why are there  $2^n$  binary strings of length  $n$ ?

$$\{0, 1\}^n = \{0, 1\} \times \{0, 1\} \times \cdots \times \{0, 1\}.$$

Product rule!  $2 \cdot 2 \cdot \cdots \cdot 2 = 2^n$ .

# Sum rule

**Rule:** If  $A_1, A_2, \dots, A_n$  are disjoint sets, then:

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n|.$$

Example:

- ▶ Mello Yello and Mello Yello Zero come in 8 flavors.
- ▶ Seagram's and Seagram's Diet come in 6 flavors.

Total sodas?  $8 \cdot 2 + 6 \cdot 2 = 28$ .



## Counting Passwords (15.2.4)

Valid password: (1) Sequence of 6 to 8 symbols. (2) First symbol must be a letter (either case). (3) Rest are letters or digits.

- ▶ Digit symbol:  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ . 10
- ▶ Letter:  $\{a, b, \dots, z\}$ . 26
- ▶ Letter symbol: letter  $\times$  {upper, lower}.  $26 \cdot 2 = 52$
- ▶ Letter/digit symbol: letter symbol  $\cup$  digit symbol.  
 $52 + 10 = 62$
- ▶ 6-symbol password: letter symbol  $\times$  5 letter/digit symbols.  
 $52 \cdot 62^5 = 5e+10$
- ▶ 7-symbol password: letter symbol  $\times$  6 letter/digit symbols.  
 $52 \cdot 62^6 = 3e+12$
- ▶ 8-symbol password: letter symbol  $\times$  7 letter/digit symbols.  
 $52 \cdot 62^7 = 2e+14$
- ▶ valid password: 6-symbol password  $\cup$  7-symbol password  $\cup$  8-symbol password.  $5e+10 + 3e+12 + 2e+14 = 2e+14$

## Counting assignments

I have 4 different planet postcards and 5 different people. How many different ways can I give the postcards to the people?

There's a bijection between sequences and postcard assignments.

If the five people are Jiahua, Kyran, Michael, Morley, and Will, I can give the first postcard to Kyran, the second to Will, the third to Morley, and the fourth to Will. The sequence is: (Kyran, Will, Morley, Will). Every assignment of the 4 postcards to the people becomes a sequence. Every sequence corresponds to a way to give the postcards to the people. Bijection. There are  $5^4 = 625$  sequences (product rule), so there are 625 ways of assigning the postcards to people.

In general,  $n$  people,  $k$  postcards,  $n^k$  assignments of the postcards to people.

## Generalized product rule

If we add the constraint that no person gets more than one postcard, the product rule no longer applies. Specifically, the entries in the list now depend on each other and we can't count them by just multiplying.

**Rule:** Let  $S$  be a set of length- $k$  sequences. If there are:

- ▶  $n_1$  possible first entries,
- ▶  $n_2$  possible second entries for each first entry,
- ▶  $\vdots$
- ▶  $n_k$  possible  $k$ th entries for each sequence of first  $k - 1$  entries,

then:

$$|S| = n_1 \cdot n_2 \cdot n_3 \cdot \cdots \cdot n_k.$$

## Counting assignments with restrictions

With the 5 people and 4 postcards case, any of the 5 can be given the first postcard. But, for the second postcard, we cannot reuse whoever got the first one. So, 4 choices left. For the third postcard, there are only 3 choices of people left. And, for the fourth postcard, there are only two people left.

$5 \cdot 4 \cdot 3 \cdot 2 = 120$ , which is less than the 625 unrestricted possibilities.

General form,  $n$  people,  $k < n$  postcards, no one gets more than one:  $n \cdot (n - 1) \cdot \dots \cdot (n - k + 1)$ .

## Serial numbers



“Defective” because the serial number (11180915) repeats a digit (1).

## Fraction non-defective

Assuming all 8-digit serial numbers equally likely, what fraction of bills are not defective?

How many 8-digit serial numbers?  $10^8$  by the product rule.

How many 8-digit serial numbers have no repeats (not defective)?  
 $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 1814400$  by the generalized product rule.

Fraction not defective: not-defective / total =  
 $1814400/100000000 = .018144$  or 1.8%.

## Pawn, Bishop, Knight

How many different ways can we place a pawn (P), a knight (N), and a bishop (B) on a chessboard so that no two pieces share a row or a column?

There are 8 rows and 8 columns, which we'll number 1 to 8.

For example, P:(1,2), N:(4,1), B:(6,2) is invalid because the pawn and bishop are in the same column (2). Moving B to (6,3) fixes it.

P's row and column are unrestricted:  $8 \cdot 8$ . N only has 7 remaining choices of row and 7 remaining choices of column:  $7 \cdot 7$ . Finally, B has 6 choices for each of row and column.

So,  $8 \cdot 8 \cdot 7 \cdot 7 \cdot 6 \cdot 6 = 112896$  different ways.

# Definition

**Definition:** A *permutation* of a set is a sequence consisting of the elements of the set each repeated exactly once.

$$S = \{w, x, y, z\}$$

$(w, x, y, z), (w, x, z, y), (w, y, x, z), (w, y, z, x), (w, z, x, y), (w, z, y, x),$   
 $(x, w, y, z), (x, w, z, y), (x, y, w, z), (x, y, z, w), (x, z, w, y), (x, z, y, w),$   
 $(y, x, w, z), (y, x, z, w), (y, w, x, z), (y, w, z, x), (y, z, x, w), (y, z, w, x),$   
 $(z, x, y, w), (z, x, w, y), (z, y, x, w), (z, y, w, x), (z, w, x, y), (z, w, y, x),$

How many permutations on  $n$  elements? It's like the non-repeating postcard problem where  $n = k$ :

$$n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1 = n!.$$



# Examples

- ▶ Assigning  $n$  professors  $n$  classes to teach (one each).
- ▶ Assigning  $n$  actors to  $n$  roles (one each).
- ▶ Visit all of  $n$  cities in some order.
- ▶ Seating arrangements of  $n$  people.
- ▶ Ways of shuffling  $n$  cards.